

Converting Between Rectangular and Polar Coordinates

Answer Key

Precalculus Polar and Parametric Representations

Quick Answers

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|------------------------------|----------------------|
| 1. $3, 3\sqrt{3}$ | 6. $-3\sqrt{3}, -3$ |
| 2. $5, 53.13$ | 7. $5, 233.13$ |
| 3. $-2\sqrt{2}, 2\sqrt{2}$ | 8. $4\sqrt{2}, 135$ |
| 4. $2\sqrt{2}, 135, 315$ | 9. $1, 1.73$ |
| 5. $60, 240, -2, -2\sqrt{3}$ | 10. $5, 90, 270, -5$ |
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Curriculum

Level: Precalculus Polar and Parametric Representations

HSF-IF.C.7, HSN-CN.B.4–B.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 24 tagged checks.

Common wrong answers

7. *If they got 53.13: reported the reference angle itself and left the point in quadrant I*
8. *If they got -45: typed the ratio into the arctangent key and wrote down whatever appeared*
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1. $(6, \frac{\pi}{3})$ to rectangular.

Step 1. The angle is already in radians and $\frac{\pi}{3}$ is a first quadrant angle, so both coordinates will come out positive — a check to make at the end.

Step 2. $x = r \cos \theta = 6 \cos \frac{\pi}{3} = 6 \cdot \frac{1}{2}$, so $x = 3$

Step 3. $y = r \sin \theta = 6 \sin \frac{\pi}{3} = 6 \cdot \frac{\sqrt{3}}{2}$, so $y = 3\sqrt{3}$

Check. $x^2 + y^2 = 9 + 27 = 36 = 6^2$ ✓ — the rectangular pair really is 6 units from the origin.

2. $(3, 4)$ to polar.

Step 1. $r = \sqrt{3^2 + 4^2} = \sqrt{25}$, so $r = 5$ — the familiar 3-4-5 triangle.

Step 2. $\tan \theta = \frac{4}{3}$, and the point is in quadrant I, where the arctangent key already returns the right answer.

Step 3. $\theta = \arctan \frac{4}{3} = 53.1301\dots^\circ$, so $\theta \approx 53.13^\circ$

Answer. $(5, 53.13^\circ)$. No correction was needed here, which is exactly why quadrant I lulls students into trusting the key everywhere.

3. $(4, \frac{3\pi}{4})$ to rectangular.

Step 1. $\frac{3\pi}{4}$ sits in quadrant II, so expect $x < 0$ and $y > 0$.

Step 2. $x = 4 \cos \frac{3\pi}{4} = 4 \left(-\frac{\sqrt{2}}{2}\right)$, so $x = -2\sqrt{2}$

Step 3. $y = 4 \sin \frac{3\pi}{4} = 4 \left(\frac{\sqrt{2}}{2}\right)$, so $y = 2\sqrt{2}$

Check. Signs match the quadrant, and $(-2\sqrt{2})^2 + (2\sqrt{2})^2 = 8 + 8 = 16 = 4^2 \checkmark$

4. $(-2, 2)$ to polar.

Step 1 (part a). $r = \sqrt{(-2)^2 + 2^2} = \sqrt{8}$, so $r = 2\sqrt{2}$

Step 2 (part b). $\tan \theta = \frac{2}{-2} = -1$. On one full turn the tangent takes each value twice, half a turn apart: $\theta = 135^\circ$ or 315°

Step 3 (part c). The point has $x < 0$ and $y > 0$, so it is in quadrant II. Of the two candidates only 135° lands there; 315° points into quadrant IV, at $(2, -2)$ — the same line through the origin, but the opposite ray.

Answer. $(2\sqrt{2}, 135^\circ)$.

Note: the two candidates always differ by exactly 180° , because $\tan$ has period 180° . That is the whole reason the quadrant has to be checked.

5. $\tan \theta = \sqrt{3}$, $r = 4$, third quadrant.

Step 1 (part a). $\tan \theta = \sqrt{3}$ gives $\theta = 60^\circ$ or 240° on $[0^\circ, 360^\circ)$.

Step 2 (part b). 60° is in quadrant I; 240° is in quadrant III, which is the one we were told. So $\theta = 240^\circ$.

Step 3. $x = 4 \cos 240^\circ = 4 \left(-\frac{1}{2}\right)$, so $x = -2$

Step 4. $y = 4 \sin 240^\circ = 4 \left(-\frac{\sqrt{3}}{2}\right)$, so $y = -2\sqrt{3}$

Check. Both coordinates negative, as quadrant III requires, and $4 + 12 = 16 = 4^2 \checkmark$

6. $(6, 210^\circ)$ to rectangular.

Step 1. 210° is 30° past a half turn, so it is in quadrant III with reference angle 30° ; both coordinates will be negative.

Step 2. $x = 6 \cos 210^\circ = 6 \left(-\frac{\sqrt{3}}{2}\right)$, so $x = -3\sqrt{3}$

Step 3. $y = 6 \sin 210^\circ = 6 \left(-\frac{1}{2}\right)$, so $y = -3$

Check. $27 + 9 = 36 = 6^2 \checkmark$, and in decimals the point is about $(-5.20, -3)$, which is where the plot lands.

7. $(-3, -4)$ to polar.

Step 1 (part a). $r = \sqrt{(-3)^2 + (-4)^2} = \sqrt{25}$, so $r = 5$ — the same 3-4-5 triangle as problem 2.

Step 2 (part b). $\tan \theta = \frac{-4}{-3} = \frac{4}{3}$, exactly the ratio from problem 2, because the two minus signs cancel. The reference angle is therefore the same 53.13° .

Step 3. But this point is in quadrant III, so add a half turn: $180^\circ + 53.1301\dots^\circ$, giving $\theta \approx 233.13^\circ$

The comparison asked for. r is identical and the reference angle is identical; only the quadrant differs, and it shifts the polar angle by exactly 180° . The tangent alone cannot tell these two points apart — which is why the plot is not optional.

Common wrong answer: 53.13° — the reference angle was reported as if the point were in quadrant I.

8. $(-4, 4)$ to polar — what the calculator does.

Step 1 (part a). $r = \sqrt{16 + 16} = \sqrt{32}$, so $r = 4\sqrt{2}$

Step 2 (part b). $\frac{y}{x} = \frac{4}{-4} = -1$, and the arctangent key returns -45° . The key always answers in $(-90^\circ, 90^\circ)$ — quadrants I and IV only — so it can never return a quadrant II angle no matter what is typed in.

Step 3 (part c). The point has $x < 0$ and $y > 0$: quadrant II. Add a half turn to the key's answer: $180^\circ + (-45^\circ)$, giving $\theta = 135^\circ$

Answer. $(4\sqrt{2}, 135^\circ)$, and -45° names the point $(4, -4)$, the mirror image through the origin.

Common wrong answer: -45° — the key's output was copied without adjusting for the quadrant.

9. The curve $r = 4 \cos \theta$.

Step 1 (part a). At $\theta = \frac{\pi}{3}$ the radius is $r = 4 \cos \frac{\pi}{3} = 2$. Then $x = r \cos \theta = 2 \cdot \frac{1}{2}$, so $x = 1$

Step 2. $y = r \sin \theta = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3} = 1.7320\dots$, so $y \approx 1.73$

Step 3 (part b) — manual review. *Full credit says:* multiply both sides by r to get $r^2 = 4r \cos \theta$. Now substitute $r^2 = x^2 + y^2$ and $r \cos \theta = x$:

$$x^2 + y^2 = 4x.$$

Complete the square in x : $x^2 - 4x + 4 + y^2 = 4$, that is $(x - 2)^2 + y^2 = 4$ — a circle of radius 2 centred at $(2, 0)$, passing through the origin.

Check. The point from part (a) should satisfy it: $(1 - 2)^2 + (\sqrt{3})^2 = 1 + 3 = 4$ ✓

Why multiply by r first: the conversion identities are $r^2 = x^2 + y^2$, $x = r \cos \theta$ and $y = r \sin \theta$. A bare $\cos \theta$ matches none of them; $r \cos \theta$ matches one exactly.

10. $(0, -5)$ — the case the formula cannot handle.

Step 1 (part a). $r = \sqrt{0^2 + (-5)^2}$, so $r = 5$

Step 2 (part b). Here $x = r \cos \theta = 0$ with $r \neq 0$, so $\cos \theta = 0$, giving $\theta = 90^\circ$ or 270° on one turn. The point is *below* the origin, so y must be negative: that selects 270° .

Step 3 (part c). Converting back: $x = 5 \cos 270^\circ = 0$ ✓ and $y = 5 \sin 270^\circ$, so $y = -5$ ✓

Step 4 — manual review. *Full credit says:* the formula $\tan \theta = \frac{y}{x}$ needs a division by x , and here $x = 0$, so the ratio is undefined — there is no angle whose tangent is $\frac{-5}{0}$. On the y -axis the tangent is not merely hard to invert, it does not exist. What decides the angle instead is the position: the point lies on the negative y -axis, which is the ray at 270° (equivalently -90°).

Note: the same thing happens on the positive y -axis, at 90° . A student who reaches for the arctangent key here gets an error message, not a wrong number — one of the few times the calculator warns you.